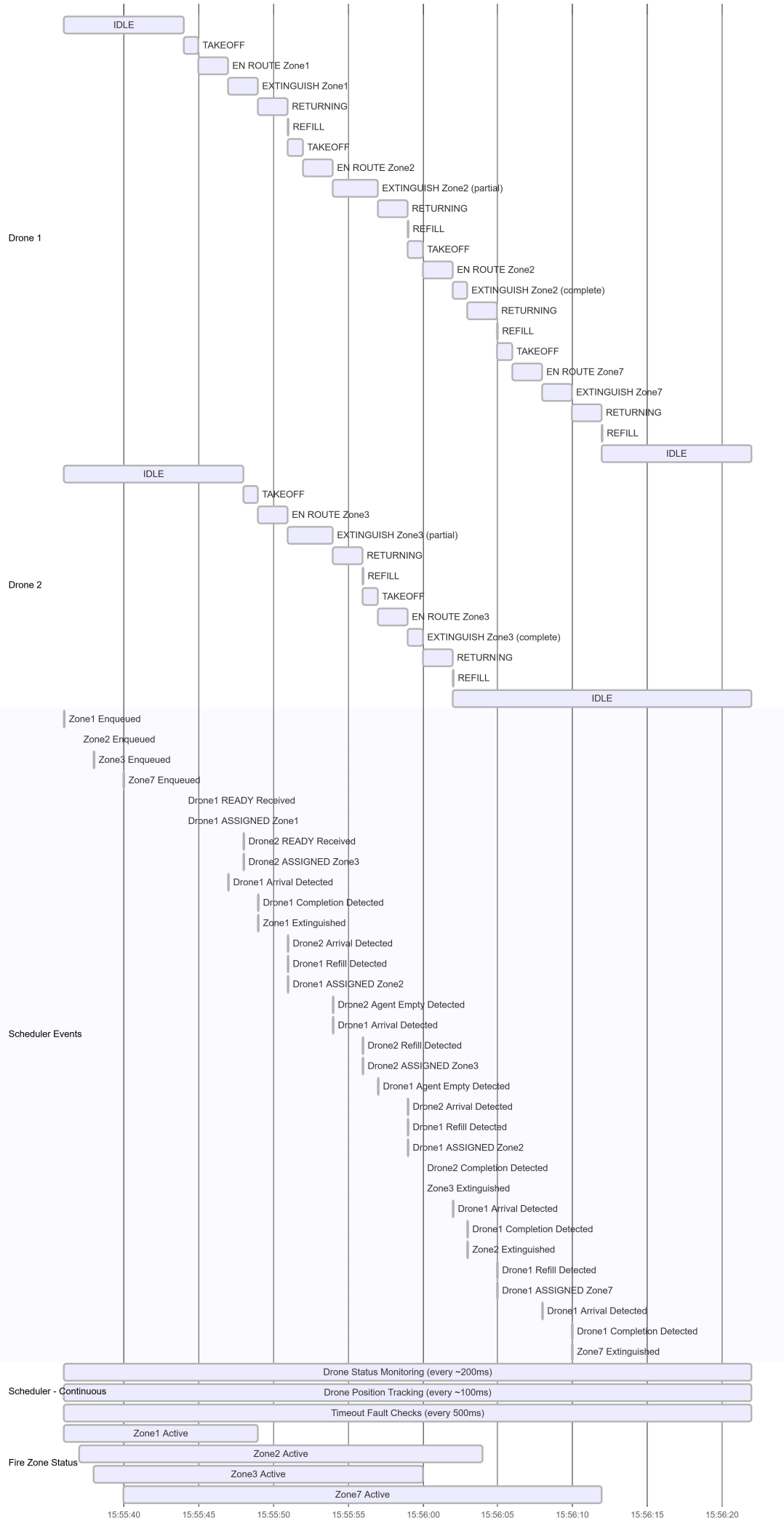
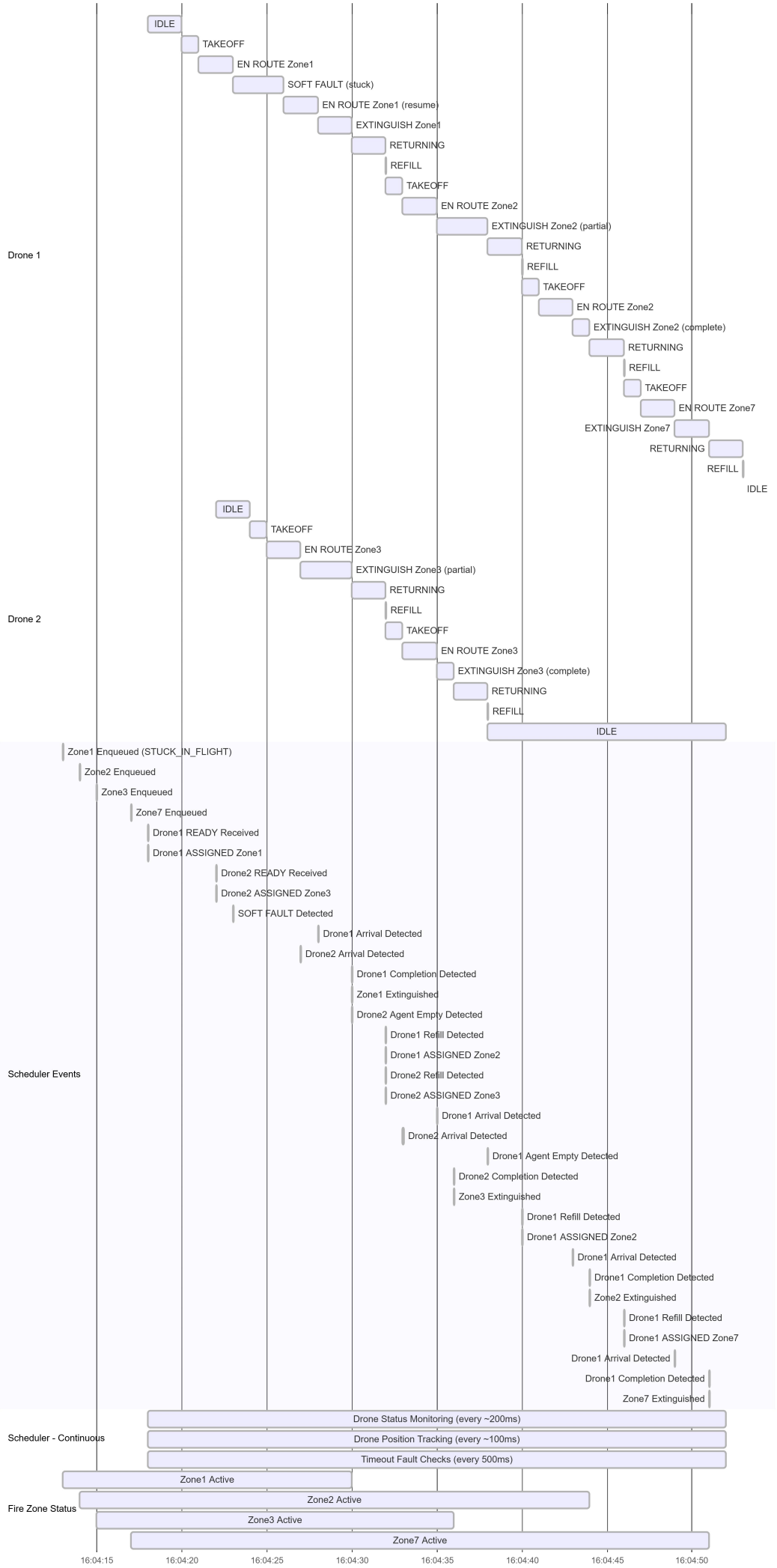
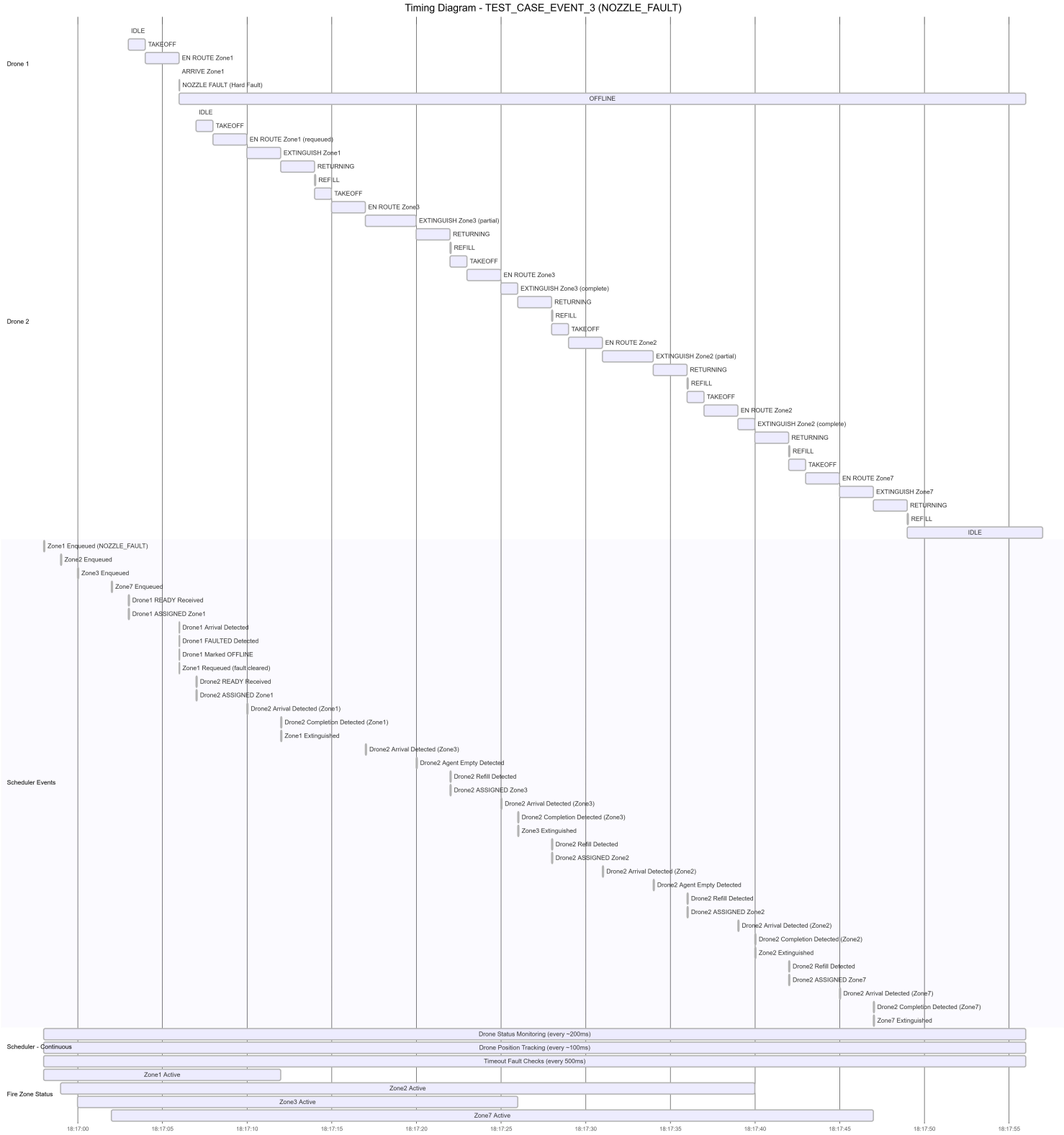
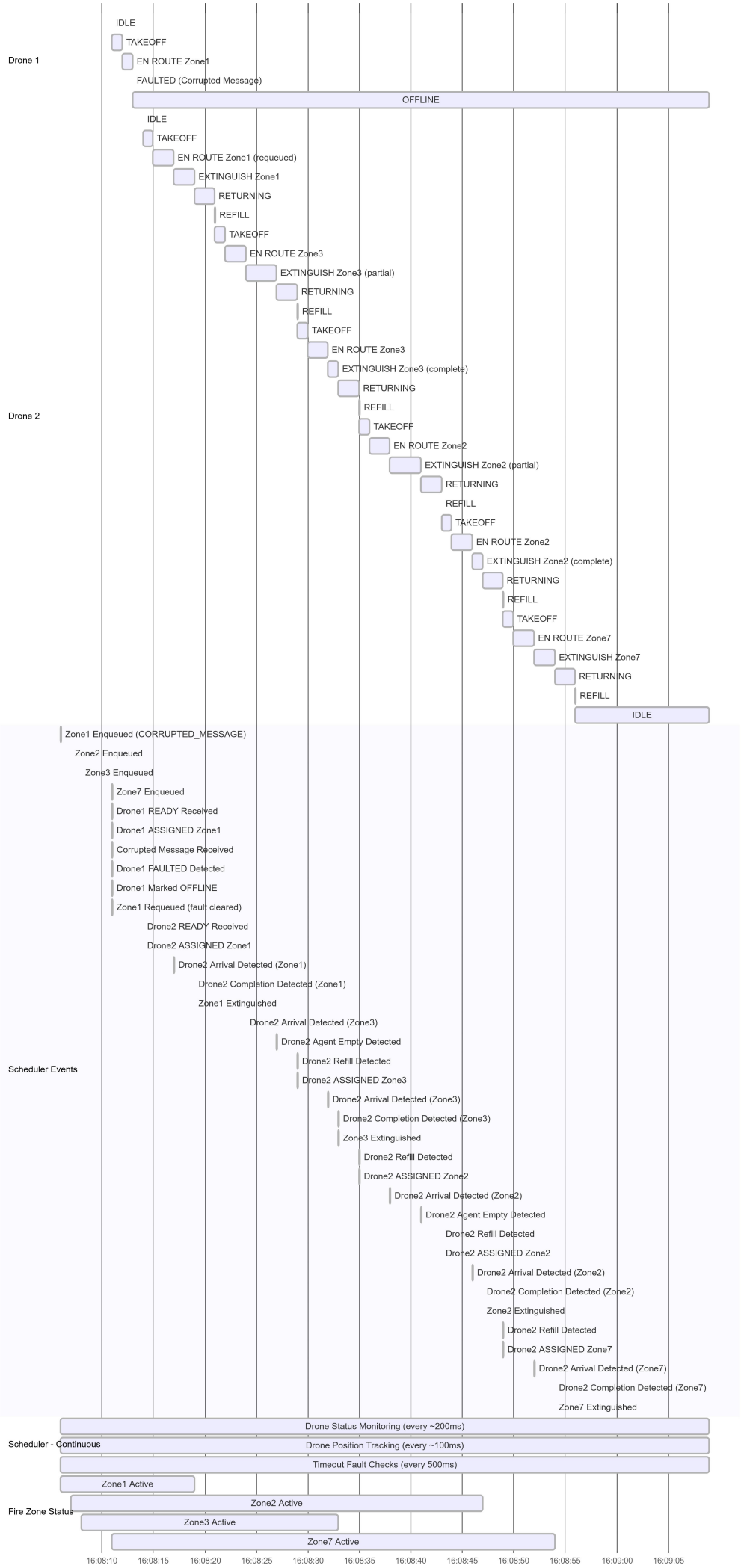
Timing Diagram - TEST_CASE_EVENT_1 (No Faults)



Timing Diagram - TEST_CASE_EVENT_2 (STUCK_IN_FLIGHT)




Timing Diagram - TEST_CASE_EVENT_3 (NOZZLE_FAULT)
Drone 1
IDLE
TAKEOFF
EN ROUTE Zone1
ARRIVE Zone1
NOZZLE FAULT (Hard Fault)
OFFLINE
IDLE
TAKEOFF
EN ROUTE Zone1 (requeued)
EXTINGUISH Zone1
RETURNING
REFILL
TAKEOFF
EN ROUTE Zone3
EXTINGUISH Zone3 (partial)
RETURNING
REFILL
TAKEOFF
EN ROUTE Zone3
EXTINGUISH Zone3 (complete)
RETURNING
REFILL
TAKEOFF
EN ROUTE Zone2
EXTINGUISH Zone2 (partial)
RETURNING
REFILL
TAKEOFF
EN ROUTE Zone2
EXTINGUISH Zone2 (complete)
RETURNING
REFILL
TAKEOFF
EN ROUTE Zone7
EXTINGUISH Zone7
RETURNING
REFILL
IDLE
Drone 2
Scheduler Events
Zone1 Enqueued (NOZZLE_FAULT)
Zone2 Enqueued
Zone3 Enqueued
Zone7 Enqueued
Drone1 READY Received
Drone1 ASSIGNED Zone1
Drone1 Arrival Detected
Drone1 FAULTED Detected
Drone1 Marked OFFLINE
Zone1 Requeued (fault cleared)
Drone2 READY Received
Drone2 ASSIGNED Zone1
Drone2 Arrival Detected (Zone1)
Drone2 Completion Detected (Zone1)
Zone1 Extinguished
Drone2 Arrival Detected (Zone3)
Drone2 Agent Empty Detected
Drone2 Refill Detected
Drone2 ASSIGNED Zone3
Drone2 Arrival Detected (Zone3)
Drone2 Completion Detected (Zone3)
Zone3 Extinguished
Drone2 Refill Detected
Drone2 ASSIGNED Zone2
Drone2 Arrival Detected (Zone2)
Drone2 Agent Empty Detected
Drone2 Refill Detected
Drone2 ASSIGNED Zone2
Drone2 Arrival Detected (Zone2)
Drone2 Completion Detected (Zone2)
Zone2 Extinguished
Drone2 Refill Detected
Drone2 ASSIGNED Zone7
Drone2 Arrival Detected (Zone7)
Drone2 Completion Detected (Zone7)
Zone7 Extinguished
Scheduler - Continuous
Drone Status Monitoring (every ~200ms)
Drone Position Tracking (every ~100ms)
Timeout Fault Checks (every 500ms)
Fire Zone Status
Zone1 Active
Zone2 Active
Zone3 Active
Zone7 Active
18:17:00
18:17:05
18:17:10
18:17:15
18:17:20
18:17:25
18:17:30
18:17:35
18:17:40
18:17:45
18:17:50
18:17:55

Timing Diagram - TEST_CASE_EVENT_4 (CORRUPTED_MESSAGE)